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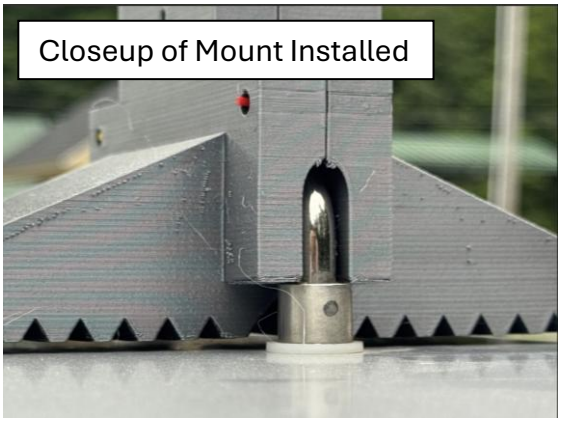
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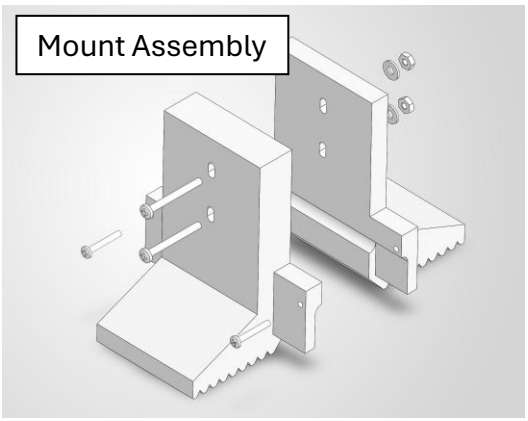
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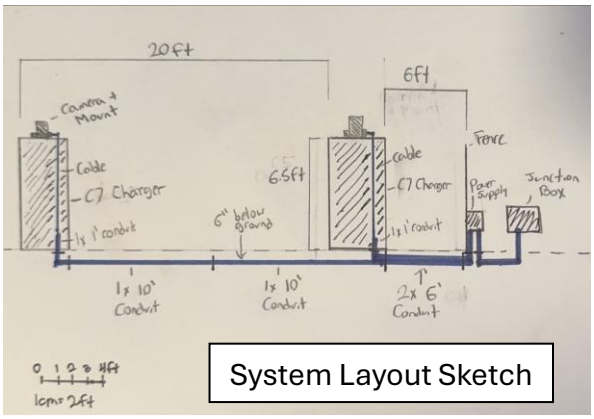
ME student graduating May 2027 seeking full-time roles in mechanical engineering.

Webcam/Intercom Retrofit – Cellerate Power









What: Designed and fabricated a retrofit for a webcam/intercom customer support and security system to be installed on top of Cellerate Power's two C7 EV chargers, without drilling into the units, which would void their warranty.

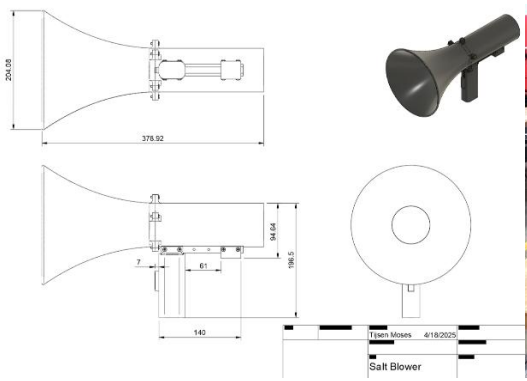
How: Took measurements at the site, and on possible webcam mounting locations. Decided that the best location to mount the webcams was on the foldable handle at the top of the chargers. The mounting hardware would have to be easily removable in case technicians need to use the handle to open the top of the unit. I chose a two-piece mount that clamps the handle together using screws pulling the two pieces together. The parts were designed in DS SolidWorks and manufactured on a 3D printer, which required four iterations of prototypes because of the handle's irregular curved geometry. To power the cameras, I had to design and install two conduits (one for each charger) with low voltage power to each unit, with an electrician making the final power connections.

Results: The system lets Cellerate see the charging area and talk to customers in real time, providing near-instant customer support when a session fails. Customer's leaving after a failed charge dropped by roughly half, and customers have given strong positive feedback regarding the way they receive help now.

Salt Distribution System – Group Project

Five-person team in Intro to Engineering Design designed and fabricated a working prototype that spreads salt evenly on walkways using a high-velocity air stream, with the salt, controls and air source carried in a backpack. Each subsystem was led by a student, and I led the Air System and Valve subsystems and designed 14+ custom components.

Air System





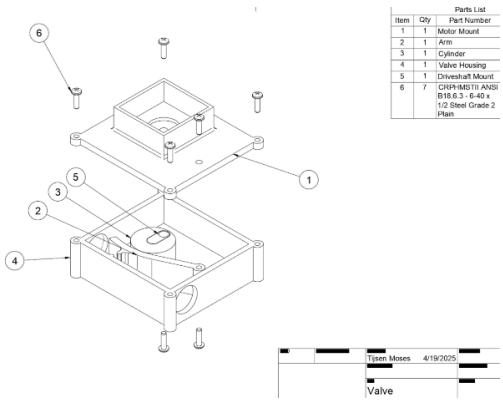
What: Designed and fabricated a handheld nozzle and tubing system to carry salt in the airflow and deposit it without blowing it away, with a handle button controlling the whole device.

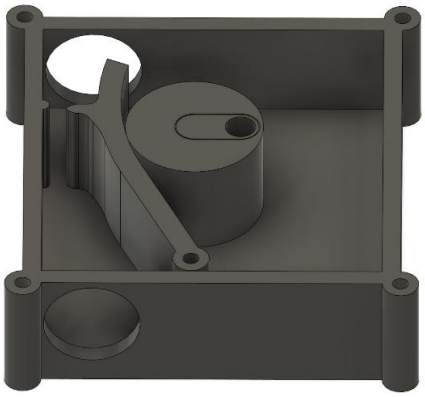
How: Used the continuity equation to size the nozzle exit so air slows from the blower rated airspeed of 150 mph in the tube to about 16 mph at the exit, slow enough that salt settles instead of blowing away. Autodesk Fusion 360 and 3D printing were used to design and manufacture the parts.

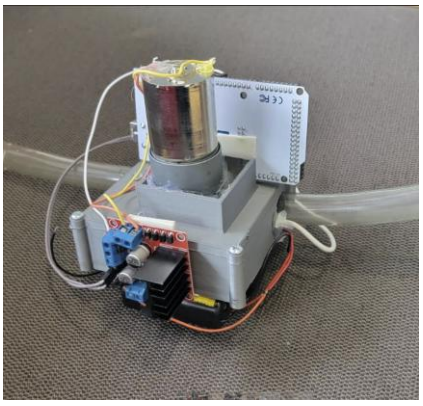
$$V_1A_1 = V_2A_2 \rightarrow V_2 = \frac{V_1\pi r_1^2}{\pi r_2^2} \rightarrow V_2 = \frac{150mph * \pi(32.5mm)^2}{\pi(100mm)\text{^}2} \rightarrow V_2 = 15.84mph$$

Results: Salt was carried reliably and landed where the nozzle was aimed, without blowing away salt already on the ground.

Valve





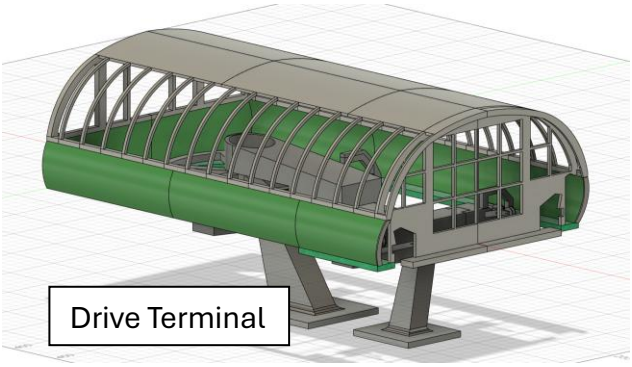


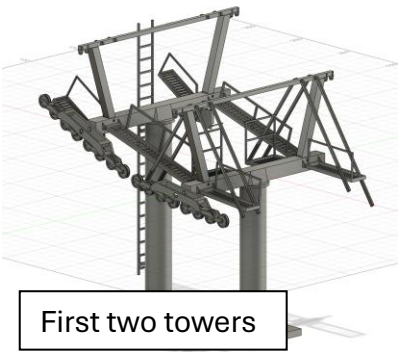
What: Designed and fabricated hardware for a valve that when integrated with electronic components, can control the release of salt into the airflow system.

How: A pinch valve type was chosen. Autodesk Fusion 360 was used to design parts that would be manufactured on 3D printers. The parts were designed together with the lead of the motor controls, to allow the motor and controls to be mounted to the valve.

Results: The valve fully sealed when closed and was able to block the flow of salt into the air system, until prompted by a button, which opened the valve and allowed salt to pass through.

Model Chairlift Project - Personal


Drive Terminal


First two towers


Chair

What: Designed and fabricated a 1/32 scale model of a Leitner-Poma “LPA” model chairlift.

How: Used Autodesk Fusion 360 to design dozens of parts and piece them together in an assembly. The drive terminal side was designed with mounting hardware for a motor and wiring. Applied 3D printer tolerance specifications to designs to ensure proper assembly. 3D printed each part, totaling over 250 parts. Utilized plastic welding and hundreds of M1 machine screws for assembly.

Result: A working motorized model chairlift. Adjustments still need to be made to how the lift is mounted to make it more structurally stable, as it can shift during operation and cause it to become misaligned and experience the rope coming off the sheaves.


Lift built and set up